

```
import csv
import sqlite3
from collections import Counter
```

```
def insert_shipping_data(db_path):
```

```
    conn = sqlite3.connect(db_path)
```

```
    cursor = conn.cursor()
```

```
    # Helper function to get or create a product ID
```

```
    def get_or_create_product_id(product_name):
```

```
        cursor.execute(
```

```
            "SELECT id FROM product WHERE name = ?", (product_name,)
        )
```

```
        row = cursor.fetchone()
```

```
        if row:
```

```
            return row[0]
```

```
        else:
```

```
            cursor.execute(
```

```
                "INSERT INTO product (name) VALUES (?)", (product_name,)
            )
```

```
            return cursor.lastrowid
```

```
# 1. Process shipping_data_0.csv
```

```
with open("shipping_data_0.csv", mode="r", encoding="utf-8") as f:
```

```
    reader = csv.DictReader(f)
```

```
    for row in reader:
```

```
        product_id = get_or_create_product_id(row["product"].strip())
```

```
        quantity = int(row["product_quantity"])
```

```
        origin = row["origin_warehouse"].strip()
```

```
        destination = row["destination_store"].strip()
```

```
        cursor.execute(
```

```
            """
```

```
            INSERT INTO shipment (product_id, quantity, origin, destination)
```

```
            VALUES (?, ?, ?, ?)
```

```
            """
```

```
            ,
            (product_id, quantity, origin, destination),
```

```
        )
```

```
# 2. Read shipping_data_2.csv
```

```
shipment_locations = {}
```

```
with open("shipping_data_2.csv", mode="r", encoding="utf-8") as f:
```

```
    reader = csv.DictReader(f)
```

```
    for row in reader:
```

```
        shipment_id = row["shipment_identifier"].strip()
```

```
        origin = row["origin_warehouse"].strip()
```

```
        destination = row["destination_store"].strip()
```

```
        shipment_locations[shipment_id] = (origin, destination)
```

```
# 3. Read shipping_data_1.csv
```

```
shipment_products = {}
```

```
with open("shipping_data_1.csv", mode="r", encoding="utf-8") as f:
    reader = csv.DictReader(f)
    for row in reader:
        shipment_id = row["shipment_identifier"].strip()
        product_name = row["product"].strip()
```

```
        if shipment_id not in shipment_products:
            shipment_products[shipment_id] = Counter()
```

```
        shipment_products[shipment_id][product_name] += 1
```

```
# 4. Combine and insert data into database
```

```
for shipment_id, products in shipment_products.items():
```

```
    if shipment_id in shipment_locations:
```

```
        origin, destination = shipment_locations[shipment_id]
```

```
        for product_name, quantity in products.items():
```

```
            product_id = get_or_create_product_id(product_name)
```

```
            cursor.execute(
```

```
                """
```

```
                INSERT INTO shipment (product_id, quantity, origin, destination)
```

```
                VALUES (?, ?, ?, ?)
```

```
                """,
```

```
                (product_id, quantity, origin, destination),
```

```
            )
```

```
conn.commit()
```

```
conn.close()
```

```
if __name__ == "__main__":
```

```
    insert_shipping_data("shipping_db.db")
```